

Blowout Preventer (BOP) Testing & Operating Procedures

Reference: API Standard 53 / API RP 64 — Permian Basin Operations

1. BOP Stack Configuration — Typical Permian Basin Land Rig

1.1 Stack Components (Bottom to Top)

Component	Type	Bore (in)	Working Pressure (psi)
Drilling Spool	—	13-5/8	5,000
Annular BOP	Hydril GK	13-5/8	5,000
Double Ram BOP	Cameron Type U	13-5/8	5,000
Single Ram BOP	Cameron Type U	13-5/8	5,000
Bell Nipple	—	20	—

Ram configurations (bottom to top): - Lower rams: Pipe rams — 5-1/2 in drill pipe -
Upper rams: Blind/shear rams

1.2 Accumulator System

- Minimum accumulator capacity: $1.5 \times$ the volume required to close all BOP components
 - Usable fluid volume (to 200 psi above pre-charge): ≥ 42 gallons per API 53
 - Pre-charge pressure: 1,000 psi nitrogen
 - Operating pressure: 3,000 psi
 - Minimum closing time (annular): ≤ 30 seconds
 - Minimum closing time (rams): ≤ 30 seconds
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2. BOP Testing Requirements

2.1 Test Frequency

- **Initial test:** After installation, before spud
- **Routine test:** Every 14 days during drilling operations (per API 53 §5.4)
- **After any repair:** Full function and pressure test required
- **After disconnection:** Full test before reconnecting to wellbore

2.2 Pressure Test Procedure

Low-pressure test (200–300 psi) 1. Close component to be tested 2. Apply 200–300 psi from test pump 3. Hold for 5 minutes — zero pressure loss required 4. Bleed pressure; confirm zero before proceeding

High-pressure test (70% of rated WP or MAASP, whichever is lower) 1. Close component 2. Apply pressure slowly (max 500 psi/min) 3. Hold for 5 minutes — zero pressure loss required 4. For Permian Basin wells: typical high-pressure test = 3,500 psi

Annular BOP test - Test to 70% of rated working pressure - Annular elements tested with drill pipe in hole - Acceptable leak rate: zero during 5-minute hold

2.3 Function Test Procedure

1. Verify accumulator pressure $\geq 2,800$ psi before test
2. Close annular BOP — verify closure on pressure gauge
3. Open annular BOP — verify full open
4. Close lower pipe rams — verify closure
5. Open lower pipe rams
6. Close upper blind/shear rams — verify closure (NO drill string in hole)
7. Open upper blind/shear rams
8. Test choke and kill line valves — open/close cycle
9. Record all opening/closing times
10. Document in BOP test log (IADC BOP Test Report)

2.4 Acceptance Criteria

- All function tests: component must fully open and close
 - All pressure tests: zero pressure loss over 5-minute hold
 - Accumulator recharge time after full closure: ≤ 5 minutes
 - Any failure: suspend drilling, notify company man, repair before resuming
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3. Well Control Procedures

3.1 Kick Detection — Warning Signs

- Pit gain (flow check required if >1 bbl unexplained gain)

- Increased flow rate with pumps off
- Drilling break (sudden ROP increase >50%)
- Decrease in pump pressure with constant stroke rate
- String weight decrease (gas cutting mud)
- Mud weight out < mud weight in

3.2 Shut-In Procedure (Hard Shut-In — Preferred)

1. Pick up off bottom (minimum 1 stand)
2. Stop mud pumps
3. Close annular BOP (or pipe rams if annular unavailable)
4. Open HCR valve to choke manifold
5. Read and record SIDPP (Shut-In Drill Pipe Pressure)
6. Read and record SICP (Shut-In Casing Pressure)
7. Read and record pit gain volume
8. Notify company man and well control supervisor
9. Do NOT bleed pressure without authorization

3.3 Kill Methods

Driller's Method (two-circulation) - Circulation 1: Circulate kick out with original mud weight - Circulation 2: Circulate kill weight mud to surface - Use when: kick volume >10 bbl or well control supervisor preference

Wait-and-Weight Method (Engineer's Method) - Weight up mud to kill weight before circulating - Single circulation to kill well - Use when: kick volume <10 bbl, time permits

Kill mud weight calculation:

$$KMW \text{ (ppg)} = OMW + (SIDPP / (0.052 \times TVD))$$

Where: OMW = original mud weight, TVD = true vertical depth (ft)

3.4 Choke Manifold Operation

- Maintain constant BHP during kill circulation
 - Adjust choke to hold SICP constant (Driller's Method) or calculated casing pressure schedule (W&W)
 - Choke operator and pump operator must communicate every 5 strokes
 - Record casing pressure, pump pressure, and strokes every 5 minutes
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4. BOP Maintenance

4.1 Daily Inspection Checklist

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Accumulator pressure $\geq$ 2,800 psi

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- ☐ Hydraulic fluid level in accumulator unit reservoir
- ☐ No visible leaks on BOP stack, choke/kill lines
- ☐ Remote panel indicators functional
- ☐ Choke manifold valves free-moving

4.2 Weekly Maintenance

- Lubricate all BOP operating rods and pins
- Check annular element condition (inspect for cuts or extrusion)
- Test remote BOP panel from driller's console
- Verify kill/choke line valve operation

4.3 Ram Packer Replacement Criteria

- Replace if: visible cuts, tears, or extrusion beyond 1/4 inch
 - Replace if: fails to hold pressure during test
 - Replace after: any shear event
 - Typical service life: 50–75 closures or 6 months, whichever comes first
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5. Emergency Procedures

5.1 Loss of Accumulator Pressure

1. Switch to manual pump (hand pump) immediately
2. Close nearest BOP component manually
3. Notify company man
4. Do not resume drilling until accumulator repaired and recharged

5.2 BOP Failure During Kick

1. Attempt secondary closure method (pipe rams if annular fails)
2. If all BOPs fail: activate emergency disconnect (if applicable)
3. Evacuate non-essential personnel
4. Notify operator, regulatory authority (Texas RRC), and well control company
5. Activate emergency response plan

5.3 Regulatory Reporting (Texas RRC)

- Blowout or loss of well control: notify Texas RRC within 24 hours
- Form W-3 (Well Blowout Report) required within 10 days
- Contact: Texas Railroad Commission, Oil & Gas Division, (512) 463-6792